

UCC306: Fluid Mechanics

L	T	P	Cr
3	1	2 *	4.0

Course Objective: To understand basic concepts of fluid flow and their applications in solving engineering problems

Introduction: Definition of a fluid and its properties, Types of fluids

Fluid statics: Differential equation of pressure field, Pascal law, Measurement of fluid pressure, force on submerged surfaces, Buoyancy and Flotation

Fluid kinematics: Methods of describing fluid motion, Velocity and acceleration of a fluid particle, Type of fluid flows, Displacement of a fluid particle, Continuity equation, Velocity potential and stream function, Flow net.

Fluid dynamics: Euler's equation; Bernoulli's equation; Momentum equation; Applications

Flow through pipes: Energy losses, Hydraulic gradient line and total energy line, Concept of equivalent pipe, Pipes in series and parallel, Flow through a siphon, Transmission of power

Flow measuring devices: Venturimeter, Orificemeter, orifice, Pitot tube, Notch, Time of emptying tanks of different cross-sections.

Dimensional analysis: Methods of dimensional analysis, Model studies

Application of computational techniques in fluid mechanics problems

Laboratory Work

1. Verification of Bernoulli's Theorem
2. Calibration of venturimeter/orifice meter.
3. Determination of hydrostatic force and its location on a vertically immersed surface.
4. Determination of friction factor for pipes of different materials
5. Determination of hydraulic coefficients of an orifice
6. Verification of momentum equation
7. Visualization of laminar and turbulent flow.

Course Learning Objectives (CLO)

The students will be able to:

1. Analyze and solve problems of simple fluid based engineering systems including pressures and forces on submerged surfaces.
2. Analyze fluid flow problems with the application of the mass, momentum and energy equations
3. Evaluate practical problems associated with pipe flow systems
4. Analyze the significant variables in hydraulic problems and to predict the performance of hydraulic prototypes.

Text Books

1. Fluid Mechanics, *Streeter*, V.L., *Wylie* E. B. and *Bedford*, K.W., McGraw Hill Book Company, 9th Edition, 2017.
2. Fluid Mechanics and Fluid Power Engineering, *Kumar D.S.*, *S. K. Kataria*, 9th Edition, 2015.

Reference Books

1. Theory and Application of Fluid Mechanics, *Subramanya*, K., Tata McGraw Hill, 1st Edition, 2010.
2. Hydraulics and Fluid Mechanics *Modi* P.N. and *Seth* S.M., Standard Book House, 22nd Edition, 2019.
3. Mechanics of Fluid *Shames* I. H., McGraw Hill, 4th Edition, 2002.
4. Introduction to Fluid Mechanics, *Fox*, R.W. and *McDonald* A.T., John Wiley and Sons. 10th Edition, 2020

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25
2	EST	40
3	Sessional: (May include the following) Assignment, Regular Lab assessmentand Quizzes, Project (Including report, presentation etc.)	35